

PERT

Michel Bierlaire

Solution of the practice quiz

1. The PERT network is defined in the following way.
 - A node is associated with each task,
 - an artificial node o is associated with the beginning of the project,
 - an artificial node d is associated with the end of the project.

The set of arcs is defined as:

- for each task j , we add an arc (i, j) for each predecessor i ,
- for each task j without predecessor, we define an arc (o, j) ,
- for each task i without successor, we define an arc (i, d) .

A cost is associated with each arc (i, j) . It is the duration of task i , irrespectively of j . The cost of each arc of the form (o, j) is zero.

We obtain the following network:

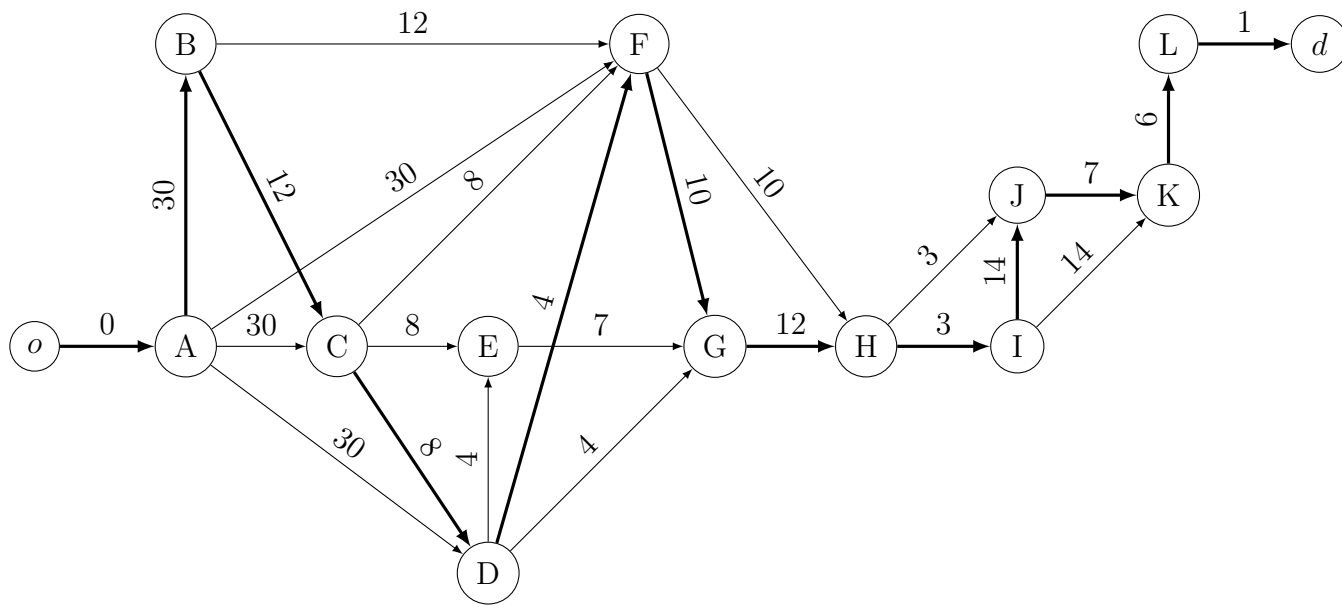


Figure 1: PERT network

2. The critical tasks are found using the longest path from o to d in the network. As the network has no cycle, we apply the shortest path algorithm on the network where all costs have the opposite sign. Indeed, even if all arcs have a negative cost, there is no cycle with negative cost.

The iterations of the algorithm are reported in Table 1. The values of π , capturing the preceding nodes in the shortest paths, are reported in Table 2

It.	S	i	λ_o	λ_A	λ_B	λ_C	λ_D	λ_E	λ_F	λ_G	λ_H	λ_I	λ_J	λ_K	λ_L	λ_d
0	$\{o\}$	o	0	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
1	$\{A\}$	A	0	0	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
2	$\{B, C, D\}$	B	0	0	-30	-30	-30	∞	∞	∞	∞	∞	∞	∞	∞	∞
3	$\{C, D, F\}$	C	0	0	-30	-42	-30	∞	-42	∞	∞	∞	∞	∞	∞	∞
4	$\{D, E, F\}$	D	0	0	-30	-42	-50	-50	-50	∞	∞	∞	∞	∞	∞	∞
5	$\{E, F, G\}$	F	0	0	-30	-42	-50	-54	-54	-54	∞	∞	∞	∞	∞	∞
6	$\{E, G, H\}$	G	0	0	-30	-42	-50	-54	-54	-64	-64	∞	∞	∞	∞	∞
7	$\{E, H\}$	H	0	0	-30	-42	-50	-54	-54	-64	-76	∞	∞	∞	∞	∞
8	$\{E, I, J\}$	I	0	0	-30	-42	-50	-54	-54	-64	-76	-79	-79	∞	∞	∞
9	$\{E, J, K\}$	J	0	0	-30	-42	-50	-54	-54	-64	-76	-79	-93	-93	∞	∞
10	$\{E, K\}$	K	0	0	-30	-42	-50	-54	-54	-64	-76	-79	-93	-100	∞	∞
11	$\{E, L\}$	L	0	0	-30	-42	-50	-54	-54	-64	-76	-79	-93	-100	-106	∞
12	$\{E, d\}$	d	0	0	-30	-42	-50	-54	-54	-64	-76	-79	-93	-100	-106	-107
13	$\{E\}$	E	0	0	-30	-42	-50	-54	-54	-64	-76	-79	-93	-100	-106	-107
13	$\{\}$	-	0	0	-30	-42	-50	-54	-54	-64	-76	-79	-93	-100	-106	-107

Table 1: Iterations of the shortest path algorithm in order to find the longest path from o to d .

It.	π_o	π_A	π_B	π_C	π_D	π_E	π_F	π_G	π_H	π_I	π_J	π_K	π_L	π_d
0	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
1	-1	o	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
2	-1	o	A	A	A	-1	-1	-1	-1	-1	-1	-1	-1	-1
3	-1	o	A	B	A	-1	B	-1	-1	-1	-1	-1	-1	-1
4	-1	o	A	B	C	C	C	-1	-1	-1	-1	-1	-1	-1
5	-1	o	A	B	C	D	D	D	-1	-1	-1	-1	-1	-1
6	-1	o	A	B	C	D	D	E	-1	-1	-1	-1	-1	-1
7	-1	o	A	B	C	D	D	E	G	-1	-1	-1	-1	-1
8	-1	o	A	B	C	D	D	E	G	H	H	-1	-1	-1
9	-1	o	A	B	C	D	D	E	G	H	I	I	-1	-1
10	-1	o	A	B	C	D	D	E	G	H	I	J	K	-1
11	-1	o	A	B	C	D	D	E	G	H	I	J	K	L

Table 2: Values of π at each iteration of the shortest path algorithm

The critical path is found by starting with node d , and proceeding backward. Its predecessor is $\pi_d = L$. The predecessor of L is $\pi_L = K$, and so on until node o is reached.

The longest path from o to d is represented in bold in the above network. Each node along that path corresponds to a critical tasks. The optimal labels of the algorithm (with the opposite sign) correspond to the longest path from the origin to the node. It corresponds to the earliest starting time of each tasks. For instance, $\lambda_D = -50$ means that task D will not start before day 50. Indeed, tasks A (30 days), B (12 days) and C (8 days) must be completed before task D can start. And it is not possible for the project to finish before $-\lambda_d = 107$ days.

3. In order to obtain the latest start for each task, we need to look at the problem in the other direction. Indeed, the “latest start” is the time the closest from the end of the project. Therefore, we reverse the directions of each arc, to obtain the network in Figure 2, and we apply the shortest path algorithm while the costs of the arcs stay unchanged. The critical path is the same, but the labels on each node tells us how much time before the end should the task start at the latest. The

optimal labels are

$$\begin{array}{ll}\lambda_A = 107, & \lambda_B = 77, \\ \lambda_C = 65, & \lambda_D = 57, \\ \lambda_E = 50, & \lambda_F = 53, \\ \lambda_G = 43, & \lambda_H = 31, \\ \lambda_I = 28, & \lambda_J = 14, \\ \lambda_K = 7, & \lambda_L = 1, \\ \lambda_d = 0.\end{array}$$

The latest starting time of task E is 50 days before the end of the project, that is $107-50=57$ days after the beginning of the project.

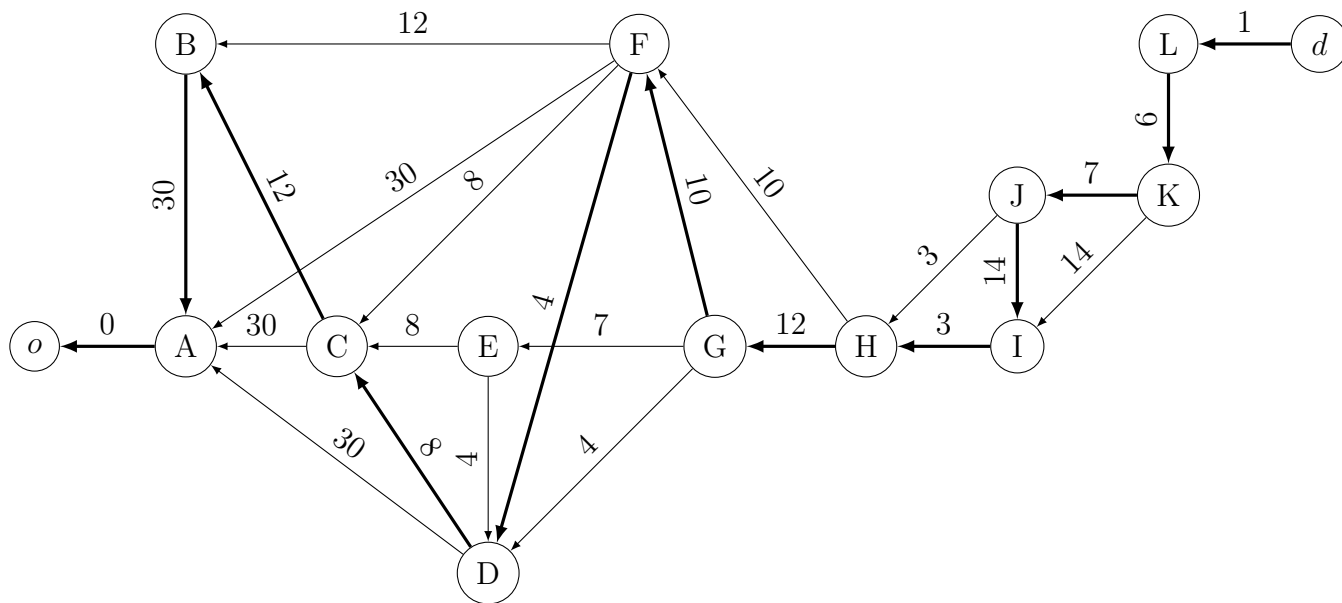


Figure 2: Reverse PERT network

The results are summarized in the table below. The project cannot last less than 107 days. The critical tasks are such that the earliest and the latest time are the same. For the others, there is some flexibility. If the task starts any time between the two dates, the project will not be delayed. In this example, the only non critical task is Task E. It cannot start before day 54, and must not start after day 57.

Ranking	Task	Duration	Precedence	Successor	Earliest start δ_i	Latest start φ_i
1	<i>o</i>	0	-	A	0	0
2	A	30	<i>o</i>	B, C, D, F	0	0
3	B	12	A	C, F	30	30
4	C	8	A, B	D, E, F	42	42
5	D	4	A, C	E, F, G	50	50
6	E	7	C, D	G	54	57
7	F	10	A, B, C, D	G, H	54	54
8	G	12	D, E, F	H	64	64
9	H	3	F, G	I, J	76	76
10	I	14	H	J, K	79	79
11	J	7	H, I	K	93	93
12	K	6	I, J	L	100	100
13	L	1	K	<i>d</i>	106	106
14	<i>d</i>	0	K	-	107	107